$\mathbb{R}^2$



$\mathbb{C} \times \mathbb{R} \longrightarrow \mathbb{R}$

$(\alpha, x) \longrightarrow \alpha * x$

$1 * x = x \Rightarrow (1+1) * x = 1 * x + 1 * x$

$2 * x = 2x$

$\Rightarrow n * x = nx \quad \cdot \quad \mathbb{N}$

$\underbrace{(1+(-1))}_{0} * x = x + (-1) * x$

$0 = x + (-1) * x$

$\Rightarrow (-1) * x = -x$

$(-n) * x = -nx \quad \mathbb{Z}$

$x + \underbrace{\left(\frac{1}{n} + \dots + \frac{1}{n}\right)}_{n} * x = n\left(\frac{1}{n} * x\right)$

$\Rightarrow \frac{1}{n} + x = \frac{1}{n} x$

$\Rightarrow \frac{m}{n} + x = \frac{m}{n} x \qquad \mathbb{Q}$

$\overset{?}{=}$

$\longrightarrow \alpha + x \overset{?}{=} \alpha x \qquad \alpha \in \mathbb{R}$

$\alpha + x$

k corp

k^n , $\mathcal{M}_{m,n}(k)$, $k[x]$

$U \subseteq V$ subspatiu $\Leftrightarrow \begin{cases} \forall\, v, w \in U \quad \forall\, \lambda, \mu \in k \\ \Rightarrow \lambda v + \mu w \in U \end{cases}$

$\Leftrightarrow \begin{cases} \forall\, v, w \in U \Rightarrow v + w \in U \\ \forall\, v \in U,\ \lambda \in k \Rightarrow \lambda v \in U \end{cases}$

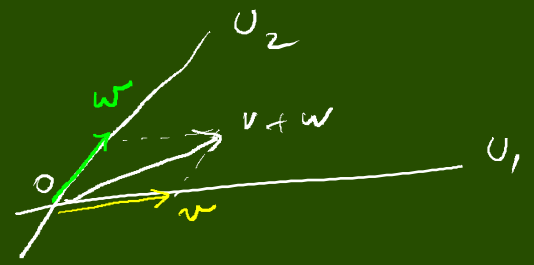
$U_1, U_2 \subseteq V$ subsp. $\Rightarrow$ $U_1 \cap U_2 \subseteq V$ DA (subspatiu?)

$U_1 \cup U_2 \subseteq V$ NU (sau)

$U_1 \cap U_2$ (obs : $U_1 \cap U_2 \neq \emptyset$!)

$0 \in U_1 \cap U_2$

$U_1 \cup U_2$



es : $U_1 \cup U_2$ subspatiu $\Leftrightarrow$?

$\Leftarrow$ ~~$\Rightarrow$~~ $U_1 = U_2$

$\Leftrightarrow$ $U_1 \subseteq U_2$ sau $U_2 \subseteq U_1$

A, B, C mulțimi

$A \subseteq C$ și $B \subseteq C \Leftrightarrow A \cup B \subseteq C$

?

Exemplu. $V = \mathbb{R}^3$ $k = \mathbb{R}$

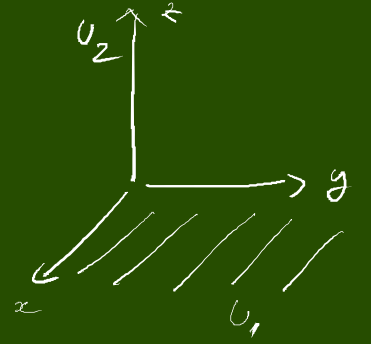
$U_1 = \{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 \mid z = 0 \}$

$U_2 = \{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 \mid x = y = 0 \}$

$U_1 + U_2 = \mathbb{R}^3$

"$\subseteq$"

"$\supseteq$" Fie $v \in \mathbb{R}^3$ $v = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$



$$v = \begin{pmatrix} x \\ y \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ z \end{pmatrix} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

U_1 U_2

$\Rightarrow U_1 + U_2 = \mathbb{R}^3$

$U_1 \cap U_2 = \{0\}$

$\Rightarrow U_1 \oplus U_2 = \mathbb{R}^3$

$\Rightarrow U_1, U_2$ complementare.

temă: determinați un subsp $W \subseteq \mathbb{R}^3$ a.î. $W \neq U_2$

$W \cap U_1 = \{0\}$

$W + U_1 = \mathbb{R}^3$ $\left(W \oplus U_1 = \mathbb{R}^3 \right)$

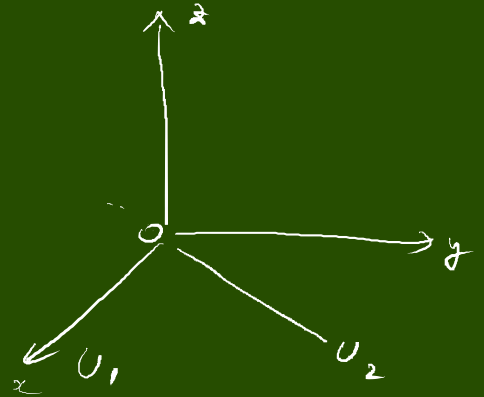
$$U_1 = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 \;/\; z = 0 \right\}$$

$$U_2 = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 \;/\; x = y,\ z = 0 \right\}$$

$U_1 + U_2 = ?$

$U_2 \subseteq U_1$

$U_1 \subseteq U_1$ $\Big/ \Rightarrow U_1 + U_2 \subseteq U_1$



$U_1 \subseteq U_1 + U_2$

$\Rightarrow U_1 + U_2 = U_1$

$$A = \begin{pmatrix} 1 & 2 & 0 \\ -3 & 0 & -2 \\ 2 & 1 & 1 \end{pmatrix}$$

$\ker(A) = ?$

$\operatorname{Im}(A) = ?$

$$\ker(A) = \{ X \in \mathbb{R}^3 \mid AX = 0_3 \}$$

$(3,3)(3,1) \quad (3,1)$

$$X = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$AX = 0_3 \Leftrightarrow \begin{cases} x + 2y = 0 \\ -3x - 2z = 0 \\ 2x + y + z = 0 \end{cases}$$

$$\left(\begin{array}{ccc|c} 1 & 2 & 0 & a \\ -3 & 0 & -2 & b \\ 2 & 1 & 1 & c \end{array}\right) \xrightarrow[L_3 - 2L_1]{L_2 + 3L_1} \left(\begin{array}{ccc|c} 1 & 2 & 0 & a \\ 0 & 6 & -2 & b+3a \\ 0 & -3 & 1 & c-2a \end{array}\right)$$

$$\xrightarrow{\frac{1}{2}L_2} \left(\begin{array}{ccc|c} 1 & 2 & 0 & a \\ 0 & 3 & -1 & \frac{b+3a}{2} \\ 0 & -3 & 1 & c-2a \end{array}\right) \xrightarrow{L_3 + L_2}$$

$$\left(\begin{array}{ccc|c} 1 & 2 & 0 & a \\ 0 & 3 & -1 & \frac{b+3a}{2} \\ 0 & 0 & 0 & c-2a+\frac{b+3a}{2} = \frac{-a+b+2c}{2} \end{array}\right)$$

$$\begin{cases} x + 2y = 0 \\ 3y - z = 0 \end{cases} \Rightarrow \begin{array}{l} x = -\frac{2z}{3} \\ y = \frac{z}{3} \end{array}$$

$z \in \mathbb{R}$

$$\underline{\ker(A)} = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 \;\middle|\; x = -\frac{2z}{3}, y = \frac{z}{3} \right\}$$

$$= \left\{ \frac{z}{3} \begin{pmatrix} -2 \\ 1 \\ 3 \end{pmatrix} \;\middle|\; z \in \mathbb{R} \right\}$$ dreapta

$$\mathrm{Im}(A) = \{ B \in \mathbb{R}^3 \mid \exists X \in \mathbb{R}^3 : AX = B \}$$

$$B = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

$$\begin{cases} x + 2y = a \\ -3x - 2z = b \\ 2x + y + z = c \end{cases} \Rightarrow \begin{cases} x + 2y = a \\ 3y - z = \frac{b + 3a}{2} \\ 0 = \frac{-a + b + 2c}{2} \end{cases}$$

$$\mathrm{Im}(A) = \left\{ B = \begin{pmatrix} a \\ b \\ c \end{pmatrix} \in \mathbb{R}^3 \mid -a + b + 2c = 0 \right\}$$

plan

$$B \in \mathrm{Im}(A) \Leftrightarrow B = \begin{pmatrix} b + 2c \\ b \\ c \end{pmatrix}, \quad b, c \in \mathbb{R}$$

$$= \begin{pmatrix} b \\ b \\ 0 \end{pmatrix} + \begin{pmatrix} 2c \\ 0 \\ c \end{pmatrix}$$

$$= b \cdot \underbrace{\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}}_{v_1} + c \cdot \underbrace{\begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}}_{v_2}$$

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ec. planului determinat de 2 vectori v_1, v_2 și de un punct $O(0,0,0)$

$v_i = \begin{pmatrix} \alpha_i \\ \beta_i \\ \gamma_i \end{pmatrix}$ $\quad i = 1, 2$

$n = v_1 \times v_2 = (\alpha_1 \bar{i} + \beta_1 \bar{j} + \gamma_1 \bar{k}) \times (\alpha_2 \bar{i} + \beta_2 \bar{j} + \gamma_2 \bar{k})$

$\bar{i} \times \bar{i} = 0$

$\bar{i} \times \bar{j} = \bar{k} = -\bar{j} \times \bar{i}$

$n = (\bar{i} + \bar{j}) \times (2\bar{i} + \bar{k}) = -\bar{j} - 2\bar{k} + \bar{i}$

$1 \cdot (x - 0) + (-1) \cdot (y - 0) + (-2)(z - 0) = 0$

$x - y - 2z = 0$

$\mathrm{Im}(A) = \{ Ax \mid x \in \mathbb{R}^3 \} = \left\{ A \begin{pmatrix} x \\ y \\ z \end{pmatrix} \mid x, y, z \in \mathbb{R} \right\}$

$A \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x + 2y \\ -3x - 2z \\ 2z + y + z \end{pmatrix} = x \begin{pmatrix} 1 \\ -3 \\ 2 \end{pmatrix} + y \cdot \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} + z \begin{pmatrix} 0 \\ -2 \\ 1 \end{pmatrix}$

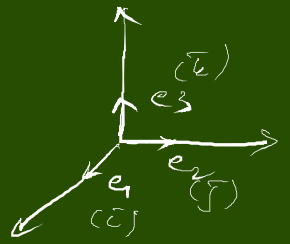
$\mathrm{Im}(A) = \{ x \cdot (\quad) + y (\quad) + z \cdot (\quad) \mid x, y, z \in \mathbb{R} \}$

$$\ker(A) = Sp\left\{\begin{pmatrix}-2\\1\\3\end{pmatrix}\right\}$$

$$Im(A) = Sp\left\{\begin{pmatrix}1\\1\\0\end{pmatrix}, \begin{pmatrix}2\\0\\1\end{pmatrix}\right\}$$

$$= Sp\left\{\begin{pmatrix}1\\-\frac{3}{2}\\\end{pmatrix}, \begin{pmatrix}2\\0\\1\end{pmatrix}, \begin{pmatrix}0\\-2\\1\end{pmatrix}\right\}$$

$$Sp\{e_1 \dots e_n\} = k^n$$



1) RGB $\begin{pmatrix}r\\g\\b\end{pmatrix}$ $\longleftrightarrow$ CMYB

2) qubit $\langle 0|$ $\langle 1|$

$q = \alpha|0\rangle + \beta|1\rangle$

ket Dirac

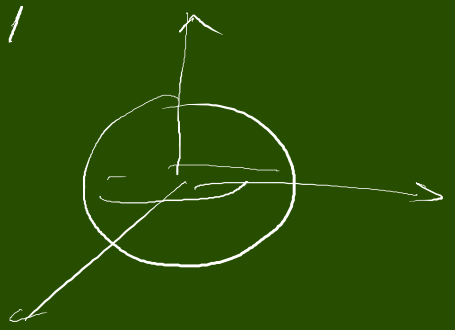
$\langle 0|$ bra

$\alpha, \beta \in \mathbb{C}$

$|\alpha|^2 + |\beta|^2 = 1$

$|0\rangle \quad |1\rangle$ braket

Bloch



$\left(v_1\right| \dots \left(v_n\right| \in k^m \longrightarrow \underbrace{\left(v_1\middle| v_2\middle| \dots \middle| v_n\right)}_{A} \in M_{m,n}(k)$

$Sp\{v_1 \dots v_n\} = \mathcal{I}m(A) \overset{?}{=} k^m$

$\mathcal{I}m(A) = \{B \in k^m \mid \exists X \in k^n,\ AX = B\}$

$A \in M_{m,n}(k)$ $\quad AX = B$ compatible

$\Leftrightarrow$ $\boxed{\operatorname{rang} A = m \quad \forall\ m \le n}$

$$\begin{cases} x + y = 1 \\ x = 1 \\ y = 1 \end{cases} \qquad A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$\Delta_{princ}^{A} \; X = B$

nec. sec. → parametri

Ex: $\mathrm{Im}(A) = \{0\}$? $A \in M_{m,n}(k)$

$\Leftrightarrow A = O_{m,n} = \begin{pmatrix} 0 & \cdots & 0 \\ \vdots & & \vdots \\ 0 & \cdots & 0 \end{pmatrix}$

$\mathrm{Im}(A) = k^m \Leftrightarrow m \le n$

$\mathrm{rang}\, A = m$

$\mathrm{Im}(I_n) = k^n$

$\ker(I_n) = \{0\}$

Def. $v_1 \dots v_n \in V$ liniar indep. $\Leftrightarrow \forall \lambda_1 \dots \lambda_n \in K$

$\lambda_1 v_1 + \dots + \lambda_n v_n = 0 \Rightarrow \lambda_1 = \dots = \lambda_n = 0$

Def. $v_1 \dots v_n \in V$ liniar dep $\Leftrightarrow \exists \lambda_1 \dots \lambda_n \in K$ cel puțin unul nenul a.î. $\lambda_1 v_1 + \dots + \lambda_n v_n = 0$

Exemple. 1) dc $v_1 = 0 \Rightarrow v_1, v_2 \dots, v_n$ liniar dep.

$\lambda_1 \neq 0$, λ_2, λ_n

$1 \cdot v_1 + 0 \cdot v_2 + \dots + 0 \cdot v_n = 0$

2) $\mathbb{R}^3$ $v_1 = \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}$ $v_2 = \begin{pmatrix} -3 \\ -6 \\ -9 \end{pmatrix}$

liniar indep.?

$3v_1 + 2v_2 = 0$ liniar dep.

$v_1 = -\frac{2}{3} v_2$ (coliniari fate)

3) $\mathbb{R}^3$

$w_1 = \begin{pmatrix} 1 \\ -3 \\ 2 \end{pmatrix}$, $w_2 = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}$, $w_3 = \begin{pmatrix} 0 \\ -2 \\ 1 \end{pmatrix}$

w_1, w_2, w_3 linear dependent.

Se $\lambda_1, \lambda_2, \lambda_3 \in \mathbb{R}$ a.i $\lambda_1 w_1 + \lambda_2 w_2 + \lambda_3 w_3 = 0$

$$\Rightarrow \begin{cases} \lambda_1 + 2\lambda_2 = 0 \\ -3\lambda_1 - 2\lambda_3 = 0 \\ 2\lambda_1 + \lambda_2 + \lambda_3 = 0 \end{cases} \Rightarrow \lambda_1 = -\frac{2}{3}\lambda_3$$

$$\lambda_2 = +\frac{1}{3}\lambda_3$$

$\Rightarrow \exists \lambda_1, \lambda_2, \lambda_3$ nem todos nulos a.i

$\lambda_1 w_1 + \lambda_2 w_2 + \lambda_3 w_3 = 0$

$\Rightarrow w_1, w_2, w_3$ lin. dep.